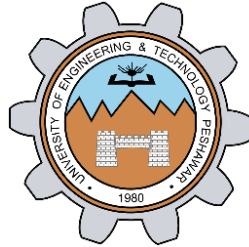


Interfacing Push buttons to 8051 Development kit using Interrupts

LAB # 04



Spring 2024

CSE-307L Microprocessor Based System Design Lab

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“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

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Date:

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Task 1:

Run the program given in the slides and attach both code and proteus Screen shot.

Code:

```
task1.c
1  #include<reg51.h>
2
3  sbit p_btn = P2^0;
4  sbit p_led = P2^1;
5  sbit i_btn = P3^2;
6  sbit i_led = P2^4;
7  int i = 0;
8
9  void e_i_0();
10
11 void main(void){
12     p_btn = 1;
13     i_btn = 1;
14     EA = 1;
15     EX0 = 1;
16     //IT0 = 1;
17     while(1){
18         if(p_btn == 0)
19             p_led = 1;
20
21         else
22             p_led = 0;
23     }
24
25 }
26 void e_i_0() interrupt 0{
27
28     for(i = 0; i<100; i++);
29     i_led ^= 1;
30 }
31
```

Output:

The screenshot displays a disassembler window titled "Disassembly" and a code editor window titled "task1.c".

Disassembly Window:

Address	Disassembly	Comment
29:	<code>i_led ^= 1;</code>	
C:0x08A4	<code>A2A4 MOV C,i_led(0xA0.4)</code>	
C:0x08A6	<code>B3 CPL C</code>	
C:0x08A7	<code>92A4 MOV i_led(0xA0.4),C</code>	
30: }		
C:0x08A9	<code>D0D0 POP PSW(0xD0)</code>	
C:0x08AB	<code>D0F0 POP ACC(0xF0)</code>	

Code Editor Window (task1.c):

```
18 if(p_btn == 0)
19     p_led = 1;
20
21 else
22     p_led = 0;
23 }
24
25 }
26 void e_i_0() interrupt 0{
27
28     for(i = 0; i<100; i++);
29     i_led ^= 1;
30 }
31 }
```

Parallel Port 3 Dialog Box:

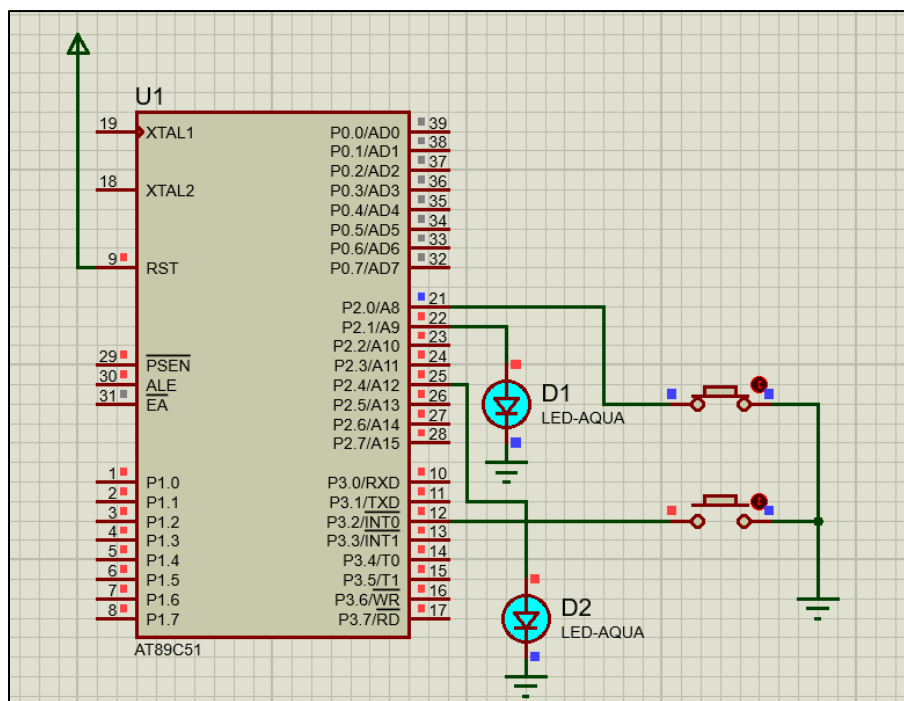
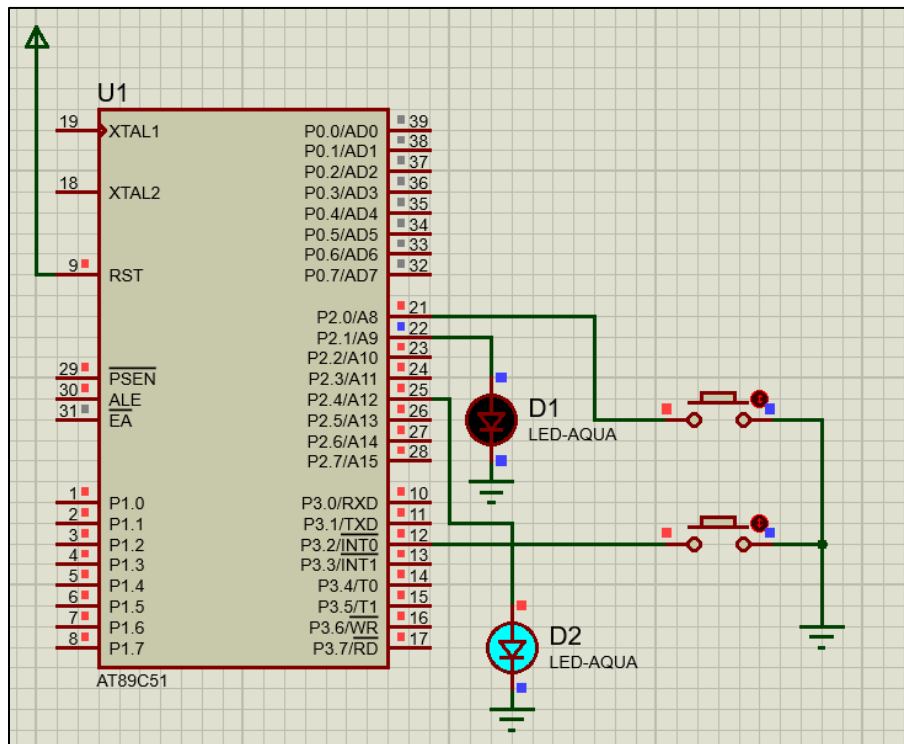
Port 3

P3: 0xFB 7 Bits 0

Pins: 0xFB

The dialog box shows a configuration for Port 3. The P3 register is set to 0xFB, and the Pins register is also set to 0xFB. The bits 7 through 0 are all checked, indicating they are configured for output.

Proteus Simulation:



Task 2:

Attach 7seg-BCD with PORT1 and whenever interrupt occur at PORT3 (P3^2) display in the seven segment. On pressing the button it display 1 again pressing the button it should display 2.

Hint: Increment PORT1 when an interrupt occur

Code:

```
task2.c
1  #include<reg51.h>
2
3  sbit i_btn = P3^2;
4
5  void e_i_0() interrupt 0{
6      P1++;
7  }
8
9  void main(void){
10
11      i_btn = 1;
12      EA = 1;
13      EX0 = 1;
14      IT0 = 1;
15
16  }
```

Output:

Disassembly

Address	Disassembly
7: }	
C:0x0817 32	RETI
C:0x0818 00	NOP
C:0x0819 00	NOP

task2.c

```
3  sbit i_btn = P3^2;
4
5  void e_i_0() interrupt 0{
6      P1++;
7  }
8
9  void main(void){
10
11      i_btn = 1;
12      EA = 1;
13      EX0 = 1;
14      IT0 = 1;
15
16  }
```

Parallel Port 1

Port 1

P1: 0x01 7 Bits 0

Pins: 0x01

Parallel Port 3

Port 3

P3: 0xFB 7 Bits 0

Pins: 0xFB

Proteus Simulation:

